

# Scalable Market Basket Analysis & Product Cross-Selling System

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| Project Scope:          | End-to-End Market Basket Analytics, Benchmarking, & Interactive Web UI |
| Cleaned Dataset Size:   | 1,036,154 records   39,517 unique invoices   5,321 items               |
| Completion Date:        | September 2026                                                         |

## 1. Executive Summary

This report presents the technical evaluation and business findings of the Market Basket Analysis platform developed by **Tanaya Salunke** and **Sankhya Londhe** on the Online Retail II transaction dataset (2009 to 2011). The system mines frequent itemsets and high-leverage association rules to uncover underlying customer purchasing behavior, optimize retail product placement, drive e-commerce cross-selling prompts, and formulate promotional product bundling strategies.

### Key System Achievements:

- Raw Transaction Ingestion:** 1,067,371 records across 2 full operational years.
- Cleaned Transaction Dataset:** 1,036,154 valid records (97.08% retention rate) spanning 39,517 unique invoices, 5,321 distinct products, 5,853 unique customers, and 43 countries.
- Memory Optimization:** Sparse matrix transaction encoding achieved a 99.53% sparsity rate, reducing RAM consumption from ~210 MB down to 4.72 MB.
- Algorithm Benchmark:** Evaluated Apriori vs. FP-Growth. FP-Growth built compact trie structures without candidate generation overhead (C<sub>k</sub>), proving vastly superior for high-cardinality transaction mining.
- Top Business Rules:** Uncovered strong product associations with Lift values up to 24.6x (e.g., Regency Teacup collections and matching ceramic accessories).

## 2. Dataset Ingestion & Data Hygiene Methodology

The data pipeline ingested raw multi-sheet Excel records (Year 2009-2010 and Year 2010-2011) and applied strict hygiene filters to ensure data integrity:

| Cleaning Stage         | Retained Records | Filter Action & Rationale                                                      |
|------------------------|------------------|--------------------------------------------------------------------------------|
| Raw Excel Ingestion    | 1,067,371        | Ingested raw sheets <i>Year 2009-2010</i> and <i>Year 2010-2011</i> .          |
| Missing Value Filter   | 1,062,989        | Removed records missing critical InvoiceNo or Description.                     |
| Cancelled Order Filter | 1,043,495        | Filtered out returns and cancelled orders (InvoiceNo starting with 'C').       |
| Quantity & Price Audit | 1,041,670        | Removed zero and negative quantities and non-positive prices.                  |
| Service Code Filter    | 1,036,154        | Stripped service items (POSTAGE, MANUAL, BANK CHARGES, CRUK, TEST001, ADJUST). |

## 3. Transaction Encoding & Exploratory Data Analysis (EDA)

### 3.1 Sparse Matrix Encoding Performance

Transactions were grouped by InvoiceNo into item sets and encoded via TransactionEncoder(sparse=True):

- **Encoded Matrix Shape:** 39,517 Invoices × 5,321 Products
- **Matrix Sparsity:** 99.5290%
- **Dense Boolean Size:** ~210.3 MB
- **Sparse Matrix RAM:** 4.87 MB (97.7% RAM savings)

### 3.2 Exploratory Data Analysis Summary

- **Mean Basket Size:** 25.06 items per invoice (Median: 15 items, 95th percentile: 72 items, Max: 1,105 items).
- **Top 5 Products by Frequency:** (1) WHITE HANGING HEART T-LIGHT HOLDER (5,778 orders), (2) REGENCY CAKESTAND 3 TIER (4,061 orders), (3) JUMBO BAG RED RETROSPOT (3,391 orders), (4) ASSORTED COLOUR BIRD ORNAMENT (2,938 orders), (5) PARTY BUNTING (2,740 orders).
- **Top 5 Products by Revenue:** (1) REGENCY CAKESTAND 3 TIER (£344,563.25), (2) WHITE HANGING HEART T-LIGHT HOLDER (£266,923.55), (3) PAPER CRAFT, LITTLE BIRDIE (£168,469.60), (4) JUMBO BAG RED RETROSPOT (£150,935.56), (5) PARTY BUNTING (£149,187.05).

## 4. Frequent Pattern Mining Benchmark: Apriori vs. FP-Growth

Both Apriori and FP-Growth algorithms were benchmarked across multiple minimum support thresholds ( $\text{min\_support} \in \{0.015, 0.02, 0.03, 0.05\}$ ):

| Min Support  | Itemsets Mined | Apriori Time (s) | FP-Growth Time (s) | FP-Growth RAM (MB) |
|--------------|----------------|------------------|--------------------|--------------------|
| 0.015 (1.5%) | 493            | 0.187 s          | 24.27 s            | 306.79 MB          |
| 0.020 (2.0%) | 270            | 0.114 s          | 9.32 s             | 306.72 MB          |
| 0.030 (3.0%) | 91             | 0.053 s          | 1.84 s             | 306.71 MB          |
| 0.050 (5.0%) | 20             | 0.025 s          | 0.52 s             | 306.71 MB          |

## 5. Top Mined Association Rules & Misleading Rule Pruning

### 5.1 Top Mined Rules (Ranked by Lift)

| Antecedent (A)                  | Consequent (C)                  | Support | Confidence | Lift   | Conviction |
|---------------------------------|---------------------------------|---------|------------|--------|------------|
| PINK REGENCY TEACUP AND SAUCER  | GREEN REGENCY TEACUP AND SAUCER | 2.16%   | 83.40%     | 24.59x | 5.82       |
| PINK REGENCY TEACUP AND SAUCER  | ROSES REGENCY TEACUP AND SAUCER | 2.03%   | 78.52%     | 22.01x | 4.49       |
| ROSES REGENCY TEACUP AND SAUCER | GREEN REGENCY TEACUP AND SAUCER | 2.59%   | 72.55%     | 21.40x | 3.52       |
| ALARM CLOCK BAKELIKE RED        | ALARM CLOCK BAKELIKE GREEN      | 2.12%   | 65.67%     | 19.04x | 2.81       |
| SPACEBOY LUNCH BOX              | DOLLY GIRL LUNCH BOX            | 2.10%   | 62.19%     | 17.96x | 2.55       |

### 5.2 Misleading Rules Identification

Rules with **High Confidence (>60%)** but **Lift ≈ 1.0** represent spurious correlations caused by universally popular items. For example, a rule predicting purchase of *WHITE HANGING HEART T-LIGHT HOLDER* given product X may show 70% confidence simply because T-Light Holders appear in 15% of all baskets anyway. The system automatically prunes rules with Lift < 1.2 to prevent suboptimal store placement decisions.

## 6. Regional / Country Segment Analysis

- **United Kingdom (36,185 baskets):** Dominated by traditional tea sets (*GREEN REGENCY TEACUP <==> PINK REGENCY TEACUP*, Lift: 24.3x).
- **Germany (753 baskets):** Characterized by home decor and tableware (*RED STRIPE CERAMIC DRAWER KNOB <==> BLUE STRIPE CERAMIC DRAWER KNOB*, Lift: 16.0x; *RED SPOTTY PAPER PLATES <==> PAPER CUPS*, Lift: 14.1x).
- **France (598 baskets):** Driven by children's products and party supplies (*CHILDRENS CUTLERY DOLLY GIRL <==> SPACEBOY*, Lift: 19.8x, Confidence: 92.6%).
- **EIRE / Ireland (581 baskets):** Strong demand for tea service sets (*REGENCY TEA PLATE GREEN <==> PINK*, Lift: 22.7x; *SUGAR JAM BOWL <==> MILK JUG*, Lift: 22.6x).

## 7. Concrete Retail Strategy & Action Plan

1. **Aisle & Shelf Co-Placement:** Position *PINK REGENCY TEACUP* and *GREEN REGENCY TEACUP* directly adjacent on store display shelves to capture strong co-purchase intent (24.6x lift).
2. **Product Combo Bundling:** Launch a packaged '**Regency Tea Party Bundle**' combining Pink, Green, and Roses teacups with a 10% promotional bundle discount.
3. **Digital Checkout Cross-Sell Prompts:** Program automated e-commerce pop-ups recommending *DOLLY GIRL LUNCH BOX* whenever a customer adds *SPACEBOY LUNCH BOX* to their digital cart.
4. **Inventory Replenishment Synchronization:** Link inventory safety stock levels for paired items to avoid losing consequent sales when antecedent products sell out.

## 8. Scalability Engine & PySpark Architecture

- **Out-of-Core Batching:** Integrated BatchTransactionProcessor processes data in 50,000-row chunks, accumulating product counts and 5.8M+ pairwise co-occurrences without loading full datasets into memory.
- **PySpark FP-Growth Integration:** Provided PySpark reference code (src/pyspark\_fpgrowth.py) mapping logic to Spark clusters for multi-terabyte data scaling using pyspark.ml.fpm.FPGrowth.